

The Stretching Machine: Squares → Parallelograms

Day 11 Enrichment · MathCraft CPA · Grade 6 · Professor Xavier Honablué, M.Ed.

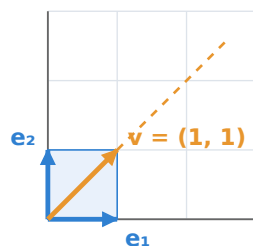
ADVANCED OUTLOOK

Advanced Outlook — not a required 6th-grade standard. This is optional enrichment for curious students. The math you use here is 6th grade (coordinates, area of rectangles, triangles, and parallelograms — 6.G.A.1, 6.G.A.3, 6.NS.C.8). The big idea behind it comes from **linear algebra**, a subject usually taught in college (MIT, Stanford, and Carnegie Mellon all teach it). Nothing on this sheet will be on a test.

The Big Idea

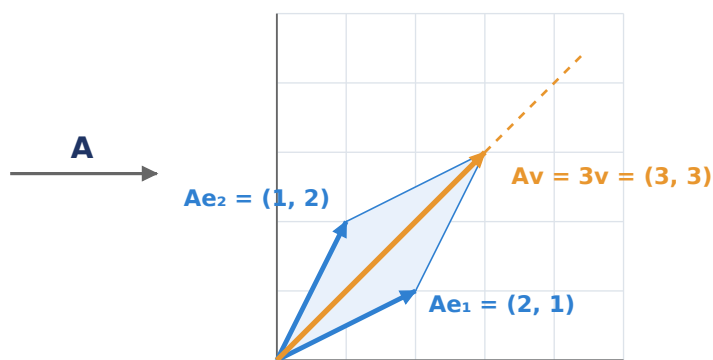
Imagine a machine called **A** that grabs the whole graph-paper grid and stretches and leans it. Every square on the grid turns into a parallelogram — and every area gets multiplied by the **same number**. Today you will find that number by using only 6th-grade tools.

Before: unit square



area 1

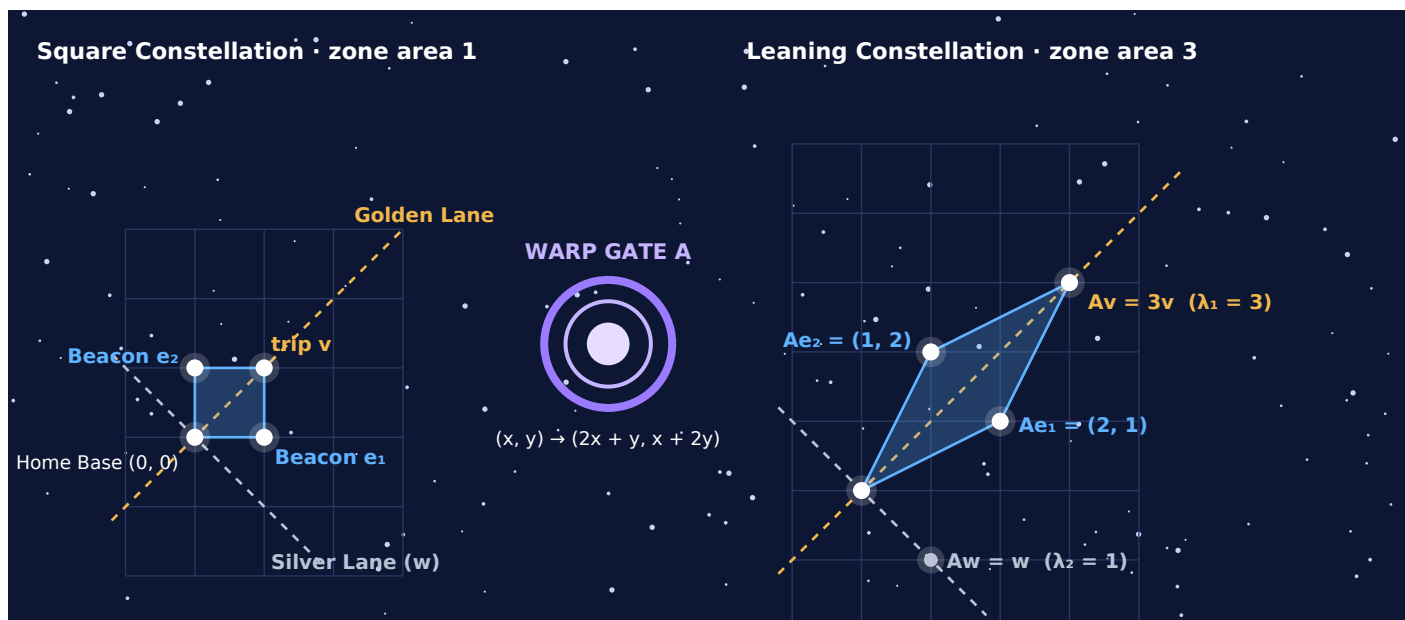
After A: parallelogram



area = $\det A = 3 = \lambda_1 \cdot \lambda_2$

Story Time · The Warp Gate of Sector A

Every symbol in the picture is part of this space story. Read it first, then use the label table as your decoder.



Chapter 1 · The Star Map

Cadet Amara is the navigator of the starship **Chandler**. Her star map is ordinary graph paper. The ship's dock, **Home Base**, sits at the corner $(0, 0)$. Each grid step is one light-year, so every small square of space is one **square light-year**.

Chapter 2 · Two Practice Beacons

Mission Control parks two beacons next to Home Base. **Beacon e_1** ("e-sub-one") is one step **east**, at $(1, 0)$. **Beacon e_2** ("e-sub-two") is one step **north**, at $(0, 1)$. Mathematicians use the letter **e** for these basic one-step arrows, and the little number underneath (the **subscript**) is just the beacon's ID number — beacon #1 and beacon #2.

Chapter 3 · The Square Constellation

Home Base, the two beacons, and one more star at $(1, 1)$ form the **Square Constellation**. The patrol zone inside it has area $1 \times 1 = \mathbf{1 \text{ square light-year}}$.

Chapter 4 · Warp Gate A

Out past the asteroid belt floats **Warp Gate A**. Space that passes through it gets stretched and leaned. The gate follows one rule for every star: $(x, y) \rightarrow (2x + y, x + 2y)$. Amara writes the gate's name in front of a beacon to mean "where it lands after the gate." So **Ae_1** means "Beacon e_1 after Gate A": $(1, 0) \rightarrow (2 \cdot 1 + 0, 1 + 2 \cdot 0) = \mathbf{(2, 1)}$. And **Ae_2** : $(0, 1) \rightarrow \mathbf{(1, 2)}$. The Square Constellation comes out as a **Leaning Constellation** — a parallelogram.

Chapter 5 · The Patrol Zone Grows

Amara uses the box method (Part 2 below) and finds the new patrol zone is **3 square light-years**. She tests a bigger square — it also comes out 3 times bigger. Gate A has an **area power of 3**: every zone that goes through gets 3 times the area. Advanced navigators call this number the **determinant** and write **$\det A = 3$** .

Chapter 6 • The Golden Lane ($\mathbf{Av} = 3\mathbf{v}$)

Captain Brooks makes one trip called $\mathbf{v}$: from Home Base to the star at (1, 1), heading northeast along the **Golden Lane**. Then she flies trip $\mathbf{v}$ through the gate: $(1, 1) \rightarrow (2 + 1, 1 + 2) = \mathbf{(3, 3)}$. Look where she lands — **still on the Golden Lane**, just 3 times as far. That is the same as making trip $\mathbf{v}$ three times in a row: $\mathbf{v} + \mathbf{v} + \mathbf{v} = 3\mathbf{v}$. The log entry $\mathbf{Av} = 3\mathbf{v}$ reads: "Gate A turned one trip $\mathbf{v}$ into three trips $\mathbf{v}$." Most routes get **bent** by the gate (Beacon $\mathbf{e_1}$ started due east but came out pointing northeast). The Golden Lane is special: the gate only **stretches** it, never bends it.

Chapter 7 • The Silver Lane

There is one more unbent route: the **Silver Lane**, heading southeast. A trip there called $\mathbf{w}$ goes to (1, -1). Through the gate: $(1, -1) \rightarrow (2 - 1, 1 - 2) = \mathbf{(1, -1)}$. The ship comes out exactly where it would have been. No bend, no stretch: $\mathbf{Aw} = \mathbf{1w}$.

Chapter 8 • Lambda, the Warp Factor (λ)

Navigators give each unbent lane a **warp factor** — how many times longer trips get on that lane. They name it with the Greek letter λ ("lambda," the Greek L — think **L** for **L**ength-changer). The subscripts again are just ID numbers: $\lambda_1 = 3$ for the Golden Lane and $\lambda_2 = 1$ for the Silver Lane.

Chapter 9 • The Captain's Secret

Captain Brooks shares a secret: "Multiply the warp factors of the unbent lanes and you get the gate's area power." $3 \times 1 = 3$. That is the whole bottom line of the picture: $\mathbf{area} = \mathbf{det\ A} = \mathbf{3} = \lambda_1 \cdot \lambda_2$ — the new area, the area power, the number 3, and warp factor times warp factor are all the same. (In college, the unbent lanes are called **eigenvectors** and the warp factors **eigenvalues**.)

Story decoder: star map = coordinate grid · Home Base = (0, 0) · beacons $\mathbf{e_1, e_2}$ = one-step arrows · Warp Gate A = the machine A · $\mathbf{Ae_1}$ = where $\mathbf{e_1}$ lands · patrol-zone area power = $\mathbf{det\ A}$ · Golden Lane = $\mathbf{v}$'s direction · $\mathbf{Av} = 3\mathbf{v}$ = trip $\mathbf{v}$ comes out 3× longer, same lane · Silver Lane = $\mathbf{w}$'s direction · warp factors $\lambda_1 = 3, \lambda_2 = 1$.

What Every Label Means

Read each label out loud the way it is written in the "Say it" column.

Label	Say it	What it means
$\mathbf{(x, y)}$	"x comma y"	An ordered pair : go x units right, then y units up from the starting corner (0, 0).
Before: unit square	"unit square"	The starting shape. Unit means 1, so it is a square with sides of 1 unit, corners at (0, 0), (1, 0), (1, 1), (0, 1).
area 1	"area one"	The unit square covers $1 \times 1 = \mathbf{1\ square\ unit}$.
$\mathbf{e_1}$	"e-sub-one"	The blue arrow pointing right , from (0, 0) to (1, 0). The small $\mathbf{1}$ is a subscript : a name tag meaning "first direction." It is NOT an exponent and NOT multiplication.
$\mathbf{e_2}$	"e-sub-two"	The blue arrow pointing up , from (0, 0) to (0, 1). The $\mathbf{2}$ means "second direction." $\mathbf{e_1}$ and $\mathbf{e_2}$ are the bottom side and left side of the unit square.
$\mathbf{v = (1, 1)}$	"vee equals one comma one"	The orange arrow . $\mathbf{v}$ is just a name for an arrow we chose to watch. It goes from (0, 0) to (1, 1) — the diagonal of the unit square.
Golden Lane / Silver Lane	—	Story names for the two unbent directions : $\mathbf{v}$'s line (1, 1) and $\mathbf{w}$'s line (1, -1). See Chapters 6–7.

Label	Say it	What it means
orange dashed line	—	The straight path that v points along, drawn longer so you can see its direction . Watch whether v stays on this line after the machine runs.
A (with the gray arrow $\rightarrow$)	"A"	The machine (math name: a transformation or matrix). Its rule: the point (x, y) moves to $(2x + y, x + 2y)$. The arrow means "apply A."
After A: parallelogram	"after A"	What the unit square becomes once A moves every point. Opposite sides are parallel and equal, so it is a parallelogram .
$Ae_1 = (2, 1)$	"A e-sub-one equals two comma one"	Writing A right in front of e_1 means " A applied to e_1 " — where the machine sends the right-arrow. Check: $(1, 0) \rightarrow (2 \cdot 1 + 0, 1 + 2 \cdot 0) = (2, 1)$.
$Ae_2 = (1, 2)$	"A e-sub-two equals one comma two"	Where A sends the up-arrow. Check: $(0, 1) \rightarrow (2 \cdot 0 + 1, 0 + 2 \cdot 1) = (1, 2)$.
light-blue shape & thin blue lines	—	The parallelogram built from Ae_1 and Ae_2 . The thin lines are copies of those two arrows that close up the other two sides. Its 4th corner is $(2, 1) + (1, 2) = (3, 3)$.
$Av = 3v$	"A vee equals three vee"	Where A sends the orange arrow: $(1, 1) \rightarrow (2 + 1, 1 + 2) = (3, 3)$. That is exactly 3 copies of v end to end — stretched 3 times longer but not turned . It stays on the dashed line.
area = det A = 3	"area equals det A equals three"	The parallelogram's area is 3 square units (you'll prove it below). det A (say "det A," short for determinant of A) is the name for the number that tells how many times bigger A makes every area. Advanced Outlook: high school / college.
$\lambda_1 \cdot \lambda_2$	"lambda-sub-one times lambda-sub-two"	λ is the Greek letter lambda . λ_1 and λ_2 are the two stretch amounts along the machine's two special directions: along v 's line everything stretches $3\times$ ($\lambda_1 = 3$); along the line through $(1, -1)$ nothing changes, $1\times$ ($\lambda_2 = 1$). The dot $\cdot$ means multiply: $3 \cdot 1 = 3$. College name: eigenvalues. Advanced Outlook.
the = signs in a chain	—	"area = det A = 3 = $\lambda_1 \cdot \lambda_2$ " says four things are all the same number: the new area, the determinant, 3, and the product of the stretch amounts.

Part 1 · Run the Machine (6th-grade skills: coordinates & expressions)

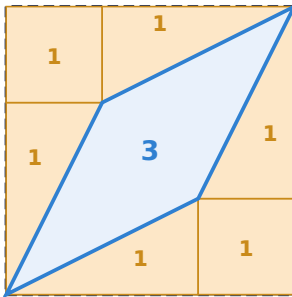
Rule of machine A: $(x, y) \rightarrow (2x + y, x + 2y)$. Fill in the table on graph paper, then plot each new point.

Start point	$2x + y$	$x + 2y$	New point
$e_1 = (1, 0)$	$2(1) + 0 = 2$	$1 + 2(0) = 1$	$(2, 1) \leftarrow$ example
$e_2 = (0, 1)$			
$v = (1, 1)$			
$(2, 0)$			
$(1, -1)$			

Parallelogram Law check: the 4th corner of the new shape should be $Ae_1 + Ae_2$. Add the x 's and the y 's: $(2 + 1, 1 + 2) = \underline{\hspace{1cm}}$. Is that the same as where v went?

Part 2 · Find the New Area with the Box Method (6.G.A.1)

The parallelogram leans, and its sides are not on grid lines, so base $\times$ height is hard to read here. Instead, **enclose and subtract** (just like Day 7 compound rectangles): draw the smallest grid box around it, then subtract the pieces outside the parallelogram.



Unit square after A

Box: $3 \times 3 = 9$

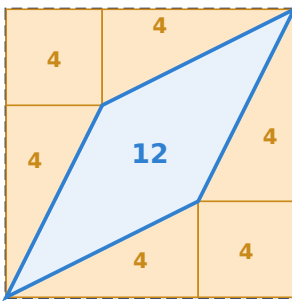
Cut-off pieces: $6 \times 1 = 6$

Parallelogram: $9 - 6 = 3$

- The box is $3 \times 3 = 9$.
- The six orange pieces are two 2-by-1 triangles (area $\frac{1}{2} \times 2 \times 1 = 1$ each), two 1-by-2 triangles (area 1 each), and two 1-by-1 squares (area 1 each): 6 in all.
- Parallelogram area = $9 - 6 = 3$ **square units**. The machine made an area-1 square into an area-3 shape.

Part 3 · Is It Always $\times 3$? Test a Bigger Square

Send the 2×2 square with corners $(0, 0)$, $(2, 0)$, $(2, 2)$, $(0, 2)$ through A. The corners land at $(0, 0)$, $(4, 2)$, $(6, 6)$, $(2, 4)$. Original area = 4. Predict: if A always triples area, the new area should be ____.



2×2 square after A

Box: $6 \times 6 = 36$

Cut-off pieces: $6 \times 4 = 24$

Parallelogram: $36 - 24 = 12$

Box $6 \times 6 = 36$, pieces $6 \times 4 = 24$, so the new area is $36 - 24 = 12 = 3 \times 4$. ✓ **Your turn:** send a 1×2 rectangle — corners $(0, 0)$, $(1, 0)$, $(1, 2)$, $(0, 2)$ — through A, use the box method, and check that the area becomes $3 \times 2 = 6$.

Part 4 · The Special Directions (Advanced Outlook)

Most arrows get **turned** by A (e_1 pointed right, but Ae_1 points up-and-right). Two directions are special:

- **Along $v = (1, 1)$:** $Av = (3, 3) = 3v$. Same direction, 3 times as long. Stretch amount $\lambda_1 = 3$.
- **Along $w = (1, -1)$:** $Aw = (2 - 1, 1 - 2) = (1, -1) = w$. Same direction, same length. Stretch amount $\lambda_2 = 1$.

Multiply the two stretch amounts: $3 \times 1 = 3$ — the same area factor you found with the box method! In college this is written **$\det A = \lambda_1 \cdot \lambda_2$** . You just discovered it with graph paper.

Beyond 6th grade (just for the curious): A is usually written as a grid of numbers called a **matrix**: $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Its determinant is $(2 \times 2) - (1 \times 1) = 4 - 1 = 3$. The special directions are called **eigenvectors** and the stretch amounts **eigenvalues**. Engineers use these ideas to design bridges and skyscrapers, and computer programmers use them for video-game graphics and search engines.

Challenge Questions

1. Warp Gate B has the rule $(x, y) \rightarrow (3x, y)$. What does B do to the unit square? What is the new area?
2. Warp Gate C has the rule $(x, y) \rightarrow (x + y, y)$. It slides the top of the square sideways (a **shear**). Plot where the unit square goes. Use base $\times$ height to find its area. Why doesn't the area change? (Hint: Day 8 — cut and slide!)
3. Gate A sends a patrol zone with area 7 square units to a new shape. What is the new area? Explain without drawing.
4. Use the box method to find the area of the parallelogram with corners $(0, 0)$, $(3, 1)$, $(4, 3)$, $(1, 2)$.

Answer Key

Part 1: $e_2 \rightarrow (1, 2)$; $v \rightarrow (3, 3)$; $(2, 0) \rightarrow (4, 2)$; $(1, -1) \rightarrow (1, -1)$. 4th corner = $(3, 3)$ — yes, the same place v went, because $v = e_1 + e_2$.

Part 3 (1 $\times$ 2 rectangle): corners go to $(0, 0)$, $(2, 1)$, $(4, 5)$, $(2, 4)$. Box $4 \times 5 = 20$. Outside pieces: bottom triangle $(0,0),(2,0),(2,1) = 1$; right piece $(2,0),(4,0),(4,5),(2,1) = 6$ (a 2-wide strip with sides 1 and 5: $2 \times (1 + 5) \div 2$); left triangle $(0,0),(0,4),(2,4) = 4$; top piece = $4 \times 1 - 1 = 3$. Total 14. Area = $20 - 14 = 6 = 3 \times 2$. ✓

1. B stretches only sideways: the unit square becomes a 3×1 rectangle, area **3**. **2.** Corners $(0, 0)$, $(1, 0)$, $(2, 1)$, $(1, 1)$: a parallelogram with base 1 and height 1, area **1**. A shear only slides pieces — no area is added or lost. **3.** A multiplies every area by 3: $7 \times 3 = 21$ square units. **4.** Box $4 \times 3 = 12$. Outside pieces: bottom triangle $(0,0),(3,0),(3,1) = 1.5$; right piece $(3,0),(4,0),(4,3),(3,1) = 2$; left triangle $(0,0),(0,2),(1,2) = 1$; top piece = $4 \times 1 - 1.5 = 2.5$. Total 7. Area = $12 - 7 = 5$. (Advanced Outlook shortcut: $3 \times 2 - 1 \times 1 = 5$.)

Explore More — Enrichment Websites

MIT, Stanford, and Carnegie Mellon resources are listed first. Items marked **Advanced Outlook** go beyond 6th grade.

MIT	MIT OpenCourseWare 18.06SC — Properties of Determinants (Prof. Gilbert Strang) Advanced Outlook · Free MIT college lecture, notes, and practice on determinants — the "det A" from our picture.
MIT	MIT OpenCourseWare 18.02 — Lecture 3 notes: determinants and area Advanced Outlook · MIT lecture notes showing how a determinant gives the area of a parallelogram.
MIT	Scratch (MIT Media Lab) 6th-grade friendly · Free block coding. Challenge: program a sprite to draw a parallelogram with the pen tool, then a sheared one with the same base and height.
STANFORD	YouCubed at Stanford — Visual Mathematics 6th-grade friendly · Stanford Graduate School of Education resources for seeing math with pictures.
STANFORD	YouCubed at Stanford — Tasks 6th-grade friendly · Free, open-ended math tasks for classes and families.

CARNEGIE MELLON	CMU CS Academy (Carnegie Mellon University) Advanced Outlook · Free graphics-based Python course from Carnegie Mellon. Draw shapes with (x, y) coordinates — try coding the Before/After picture.
OPEN · GR 6	Illustrative Mathematics — Grade 6, Unit 1, Lesson 4: Parallelograms 6th-grade standard · Decompose and rearrange parallelograms to find area.
GEOGEBRA	GeoGebra — Area of a Parallelogram (Discovery) 6th-grade standard · Drag the corners and a slider to watch a parallelogram become a rectangle.
3BLUE1BROWN	3Blue1Brown — The Determinant (lesson page) Advanced Outlook · Animated explanation of how a machine scales area — the exact idea in our picture.